

Does Offshoring Increase Domestic Employment?

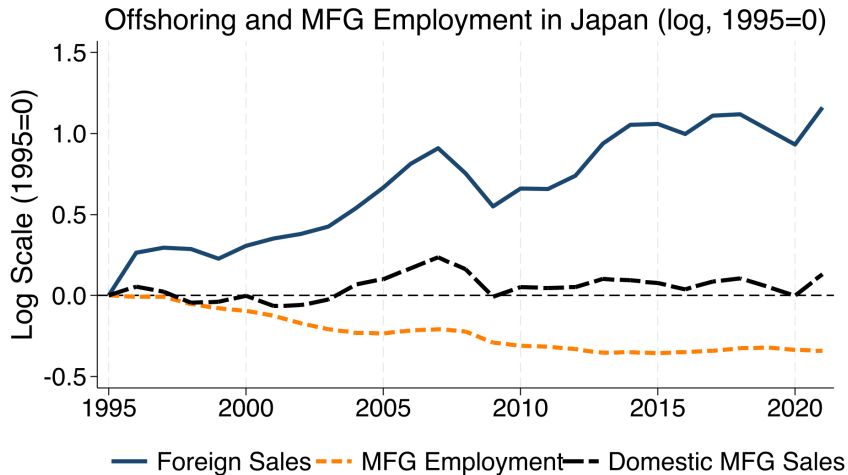
Shinnosuke Kikuchi

(joint with Ippei Fujiwara and Satoshi Ichikawa)

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Econometric Society at Hong Kong

Rise of Offshoring, Decline of Manufacturing Employment



Source: BSOBA, BSJBSA, JIP

More Offshoring, Fewer Jobs?

- Important and Controversial Topic
 - “One of the **most controversial** aspects of globalisation is offshoring.”
 - Kiyota et al (2022, VOXEU)
 - Re-shoring attempts by many governments
- “**little agreement** among academic economists regarding the sign...”
 - Kovak et al (2021, RESTAT)
- Some evidence (in the US or Japan)
 - Parents' overall emp: ↑: Kovak et al (2021, RESTAT),...
 - Inequality: ↑: Feenstra and Hanson (1999, QJE), Hummels et al (2014, AER)
 - Regional/Sectoral emp: zero or ↑? Kovak et al (2021), Kiyota et al (2025)

This Paper

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1. Across Firms – No. Employment increases.

- Revisit Kovak et al (2021) in Japan
- Confirm the key result (and the results are robust)

$$\text{corr}(\Delta\text{Offshoring}, \Delta\text{Parents' EMP}) > 0$$

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RQ: Does Offshoring **Decrease** Domestic Employment?

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$$\text{corr}(\Delta\text{Offshoring}, \Delta\text{Parents' EMP}) > 0$$

2. **Aggregate (Japan) – Yes.**

2.1 Confirm sizable substitution against domestic suppliers

$$\text{corr}(\Delta\text{Parents' Offshoring}, \Delta\text{Suppliers' EMP}) < 0$$

2.2 Simple PE accounting: Net job loss of 398K (rel 1.26M)

2.3 GE counterfactual: Still net job loss (1/3 of PE)

Firm-level Evidence

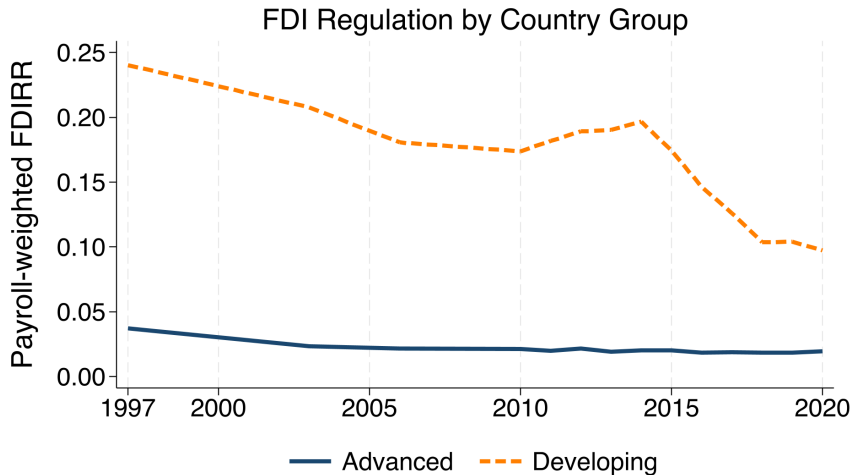
Offshoring and Employment

- Firm-level changes in employment on changes in offshoring

$$\Delta \ln L_i = \beta \Delta \text{OFS}_i + \varepsilon_i$$

- ΔOFS_i : Changes in firm i 's foreign affiliates' employment (BSOBA)
- Of course, this is problematic—Omitted Variable Bias
 - Positive demand/supply shock to the firm can increase both L and OFS

FDI Regulation Index by Country



Source: BSOBA, OECD

Firm-level Analysis: Specification

- Long-difference between 2010-2019
- Follow standard SSIV (long-difference); i firms

$$\Delta \ln L_i = \beta \Delta \text{OFS}_i + \varepsilon_i$$

- ΔOFS_i : Changes in foreign affiliates' employment (BSOBA)
- SSIV

$$\Delta Z_i \equiv \sum_{c,s} \underbrace{\frac{Y_{i,c,s,t0}}{Y_{i,t0}}}_{\text{Share}} \cdot \underbrace{\Delta Z_{c,s}}_{\text{Shift}}$$

- $\Delta Z_{c,s}$: Country-sector specific FDI Regulation Index (OECD)
- $Y_{i,c,s,t}$: Payroll of firm i ' affiliate in country c , sector s , time t (BSOBA)

1st Stage: Looser Regulations Increase FDI

Dep. Var.	Change In FDI			
	(1)	(2)	(3)	(4)
FDIRR Exposure	0.68 (0.46)	-6.31*** (1.36)	-6.23*** (1.38)	-6.36*** (1.26)
Host GDP Change		-3.40*** (0.59)	-3.42*** (0.66)	-3.38*** (0.74)
Host RER Change			0.09 (0.67)	0.25 (0.77)
China Exposure				-0.11 (0.31)
Obs.	2,241	2,241	2,241	2,241
First-Stage F	2.16	21.44	20.46	25.48

Source: BSOBA, BSJBSA

Reduced Form: Looser Regulations Increase Employment

Dep. Var.	Total Employment			
	(1)	(2)	(3)	(4)
FDIRR Exposure	-0.24*** (0.09)	-0.51*** (0.15)	-0.34* (0.20)	-0.41** (0.20)
Host GDP Change		✓	✓	✓
Host RER Change			✓	✓
China Exposure				✓
Obs.	2,241	2,241	2,241	2,241

Source: BSOBA, JIP, CoM

2SLS Results: FDI Increases Parents' Employment

Dep. Var.	Total Employment			
	(1)	(2)	(3)	(4)
Change In FDI	-0.35 (0.28)	0.08*** (0.03)	0.05 (0.03)	0.07* (0.03)
Host GDP Change		✓	✓	✓
Host RER Change			✓	✓
China Exposure				✓
Obs.	2,241	2,241	2,241	2,241
First-Stage F	2.16	21.44	20.46	25.48

Source: BSOBA, BSJBSA

Summary of Firm-level Empirical Results

- Revisit/replicate Kovak et al (2021) in Japan
- Confirm the key result (and the results are robust) across Firms

$$\text{corr}(\Delta\text{Offshoring}, \Delta\text{EMP}) > 0$$

Effects on Suppliers

Supplier-level Analysis: Specification

- Long-difference between 2010-2019
- Again, SSIV (long-difference); s supplier firms

$$\Delta \ln L_s = \beta \left[\sum_{i \in B(s)} \omega_{i,s} \cdot \Delta \text{OFS}_i \right] + \varepsilon_{s,t}$$

- SSIV: Suppliers' exposure to buyer-weighted FDI regulation

$$\Delta Z_s = \sum_{i \in B(s)} \omega_{i,s} \cdot \Delta Z_i, \quad \Delta Z_i \equiv \sum_{c,s} \frac{Y_{i,c,s,t0}}{Y_{i,t0}} \cdot \Delta Z_{c,s}$$

where $\omega_{i,s}$ is employment weight and ΔZ_i is our previous IV

2SLS: Buyers' FDI Decreases Suppliers' Employment

Dep. Var.	Change In Supplier Employment			
	(1)	(2)	(3)	(4)
Change In FDI	-0.18*** (0.03)	-0.24*** (0.03)	-0.24*** (0.03)	-0.24*** (0.03)
Host GDP Change IV		✓	✓	✓
Host RER Change IV			✓	✓
Own FDIRR Control				✓
Obs.	490,856	490,856	490,856	490,856
First-Stage F	223.19	111.92	85.55	83.87

Source: BSOBA, TSR

2SLS: Related Suppliers Survive

Dep. Var.	Change In Supplier Employment			
	(1)	(2)	(3)	(4)
Related FDI	0.63 (0.53)	0.54*** (0.20)	0.27*** (0.10)	0.27*** (0.10)
Unrelated FDI	-0.20*** (0.03)	-0.26*** (0.03)	-0.27*** (0.03)	-0.27*** (0.03)
Split Host GDP Change IVs		✓	✓	✓
Split Host RER Change IVs			✓	✓
Own FDIRR Control				✓
Obs.	490,856	490,856	490,856	490,856
F-Stat., Related FDI	5.89	6.75	7.24	7.97
F-Stat., Unrelated FDI	127.91	65.08	48.07	47.37

Source: BSOBA, TSR

Simple Accounting: 2010-2019

- Before doing a full GE, try a simple PE accounting
 - "How many jobs increase/decrease per predicted offshoring, $\widehat{\Delta OFS_i}$?"

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- Direct effect (firm/parent-level regression coef $\widehat{\beta}^P$)

$$\widehat{\Delta L}^{\text{Direct}} = \sum_i L_{i,t0} \cdot \exp \left(\widehat{\beta}^P \cdot \widehat{\Delta OFS}_i \right) \approx 121K$$

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$$\widehat{\Delta L}^{\text{Indirect}} = \sum_s L_{s,t0} \cdot \exp \left(\widehat{\beta}^S \cdot \widehat{\Delta OFS}_s \right) \approx -520K$$

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$$\widehat{\Delta L}^{\text{Indirect}} = \sum_s L_{s,t0} \cdot \exp \left(\widehat{\beta}^S \cdot \widehat{\Delta OFS}_s \right) \approx -520K$$

- Net Effect: $\widehat{\Delta L}^{\text{Direct}} - \widehat{\Delta L}^{\text{Indirect}} \approx -398K$

Quantitative Model

Model Overview

Small open economy with two sectors (tradable vs non-tradable)

1. Tradable goods producers (=buyers) use domestic and foreign intermediates
 - Offshoring = substitution between domestic and foreign inputs
2. Non-tradable goods producers + Usual across-sector demand
3. Market clearing

Model 1/3. Tradable final-good producers

- Production

$$y_i = (\ell_i)^\alpha x_i^\beta, \quad \alpha > 0, \quad \beta > 0, \quad \alpha + \beta < 1.$$

where intermediates x_i is

$$x_i = \left[\omega^{1/\sigma} \left(x_i^D \right)^{\frac{\sigma-1}{\sigma}} + (1-\omega)^{1/\sigma} \left(x_i^F \right)^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}$$

- Domestic inputs from suppliers

$$p^D = \frac{w}{z^S}.$$

- Foreign inputs available elastically at the exogenous price

$$p^F = \tau \cdot \overline{p^F},$$

where τ is an offshoring wedge.

Model 2/3. Non-Tradable and Household Demand

- Non-tradable Production

$$Y^N = z^N \ell^N,$$

- Households' demand for sectoral goods

$$U(C^T, C^N) = (C^T)^{1-\eta} (C^N)^\eta,$$

s.t.

$$, I = w\bar{L} + \Pi, \quad \Pi \equiv \sum_{i \in \mathcal{I}} \pi_i,$$

Model 3/3. Market Clearing and Equilibrium

- Domestic intermediate market $X^D = \sum_{i \in \mathcal{I}} x_i^D$.
- Nontradable market $Y^N = C^N$.
- Domestic labor market $\sum_{i \in \mathcal{I}} \ell_i + \ell^S + \ell^N = \bar{L}$.
- A competitive equilibrium consists of a wage, prices, and allocations such that
 - Producers solve profit maximization
 - Households maximize utility
 - Markets clear

Effects of Offshoring

Equilibrium Characterization 1/2: Input Substitution

Foreign input price falls with τ

$$\tau \downarrow \implies p^F \downarrow$$

Total intermediate price falls

$$d \log P^X = s_F d \log p^F < 0$$

Share of domestic inputs falls

$$d \log s_D = (\sigma - 1) s_F d \log p^F < 0$$

Therefore, when τ decreases,

$P^X \downarrow,$	$s_D \downarrow,$	$s_F = 1 - s_D \uparrow$
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Equilibrium Characterization 2/2: Substitution vs Scale

Let $r \equiv 1 - \alpha - \beta$. Then from labor demand,

$$d \log L^P = -\frac{\beta}{r} s_F d \log p^F$$

so that

$$p^F \downarrow \implies Y \uparrow, L^P \uparrow.$$

For domestic suppliers,

$$d \log L^S = \left[(\sigma - 1) - \frac{\beta}{r} \right] s_F d \log p^F.$$

Hence effects on domestic inputs suppliers will be

$$L^S \downarrow \iff \sigma - 1 > \frac{\beta}{r}.$$

Calibration: Two IV Moments, Two Elasticities

Block	Objects	Values	Discipline
Baseline	(s_F^0, q, η)	(0.113, 0.112, 0.171)	JIP/TSR; baseline clearing
IV targets	$(\theta_P^{IV}, \theta_S^{IV})$	(0.0918, -0.2355)	Parent and supplier 2SLS
Elasticities	$(\beta/r, \sigma)$	(0.811, 3.890)	Analytic IV inversion
Production	(δ, α, β)	(0.477, 0.054, 0.424)	q and β/r
Counterfactual	$\hat{\tau}_1$	0.886	Match +1.26 M affiliate jobs

Quantitative Results

Exercise.

- Choose τ_1 to reproduce FDIRR-predicted increase of 1.26 M affiliate jobs
 - $\tau : 1 \rightarrow 0.886$
- then solve the wage-clearing general equilibrium.

Results:

$$(s_F, s_D) : (11.3, 88.7)\% \rightarrow (15.0, 85.0)\%$$

$$P^x : -2.4\% \quad Y : +2.1\% \quad w : -1.0\%$$

GE Wage Dampens the Employment Loss

Employment outcome	PE accounting	GE counterfactual	GE (fixed wage)
Parent firms	+121K		
Suppliers	−520K		
Exposed sector	−398K		
Nontradables	—		
Total domestic	—		

GE Wage Dampens the Employment Loss

Employment outcome	PE accounting	GE counterfactual	GE (fixed wage)
Parent firms	+121K	+83K	
Suppliers	-520K	-221K	
Exposed sector	-398K	-138K	
Nontradables	-	+138K	
Total domestic	-	0, fixed $\bar{L}$	

GE Wage Dampens the Employment Loss

Employment outcome	PE accounting	GE counterfactual	GE (fixed wage)
Parent firms	+121K	+83K	+35K
Suppliers	-520K	-221K	-617K
Exposed sector	-398K	-138K	-582K
Nontradables	-	+138K	+582K
Total domestic	-	0, fixed $\bar{L}$	0, fixed $\bar{L}$

GE Wage Dampens the Employment Loss

Employment outcome	PE accounting	GE counterfactual	GE (fixed wage)
Parent firms	+121K	+83K	+35K
Suppliers	-520K	-221K	-617K
Exposed sector	-398K	-138K	-582K
Nontradables	-	+138K	+582K
Total domestic	-	0, fixed $\bar{L}$	0, fixed $\bar{L}$

- The exposed-sector net loss remains negative (but 1/3 of PE)
- Wage adjustment cushions supplier displacement

Conclusion

- Offshoring **decreases** (exposed sector) employment at the **aggregate level**
 - opposite to the **cross-firm** sign

$$\text{corr}(\Delta\text{Offshoring}, \Delta\text{EMP}) > 0$$

- Caution for extrapolating micro evidence to macro questions